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Transcriptome analysis of left versus right intrinsic laryngeal muscles associated with innervation

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Abstract

Objectives/Hypothesis: Recurrent laryngeal nerve injury diagnosed as idiopathic or due to short-term surgery-related intubation exhibits a higher incidence of left-sided paralysis. While this is often attributed to nerve length, it is hypothesized there are asymmetric differences in the expression of genes related to neuromuscular function that may impact reinnervation and contribute to this laterality phenomenon. To test this hypothesis, this study analyzes the transcriptome profiles of the intrinsic laryngeal muscles (ILMs), comparing gene expression in the left versus right, with particular attention to genetic pathways associated with neuromuscular function.

Study Design: Laboratory experiment.

Methods: RNA was extracted from the left and right sides of the rat posterior cricoarytenoid (PCA), lateral thyroarytenoid (LTA), and medial thyroarytenoid (MTA), respectively. After high-throughput RNA-Sequencing (RNA-Seq), 88 samples were organized into 12 datasets according to their age (P15/adult), sex (male/female), and muscle type (PCA/LTA/MTA). A comprehensive bioinformatics analysis was conducted to compare the left-right ILMs across different conditions.

Results: 774 differentially expressed genes (DEGs) were identified across the 12 experimental groups, revealing age, sex, and muscle-specific differences between the left versus right ILMs. Enrichment analysis of Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways implicated several genes with a left-right laryngeal muscle asymmetry. These genes are associated with neuronal and muscular physiology, immune/inflammatory response, and hormone control.

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Conclusion: Bioinformatics analysis confirmed divergent transcriptome profiles between the left-right ILMs. This preliminary study identifies putative gene targets that will characterize ILM laterality.

Level of Evidence: N/A

LAY SUMMARY

Vocal fold paralysis is more common on the left. This study shows left versus right differences in gene expression related to innervation, suggesting the increased rate of left recurrent laryngeal nerve paralysis may be associated with genetic differences, not just nerve length.

Keywords

Laterality; innervation; laryngeal muscles; recurrent laryngeal nerve injury; neuroinflammation; vocal fold paralysis

INTRODUCTION

Normal vocal fold function is necessary for the fundamental activities of daily life: eating, breathing, and speaking. Recurrent laryngeal nerve (RLN) injury results in vocal fold paralysis due to synkinetic reinnervation.[1] Vocal fold paralysis causes dysphagia and dysphonia, which leads to frustration, isolation, fear, lost worker productivity, and altered self-identity.[2–5] While current treatments have adequate results, they are suboptimal and disability claims are at rates similar to those of chronic diseases like congestive heart failure, chronic obstructive pulmonary disease, and severe depression.[2]

Efforts to restore normal vocal fold function focus on the guidance of reinnervating injured axons back to the correct laryngeal muscle to prevent synkinetic reinnervation and the resultant paralysis.[1–6] Understanding the genes expressed by ILMs related to innervation and reinnervation in the healthy state and after RLN injury may identify putative molecular pathways involved in guiding innervation. These genes can be investigated as therapeutic targets for manipulation after nerve injury to guide axons to the appropriate muscle and restore normal vocal fold function.

Robust clinical evidence supports a greater incidence of left vocal fold paralysis versus right.[7–15] Sidedness of the paralysis has been hypothesized to result from the longer length of the left RLN compared to right. Further, the left RLNs course through the thorax may also increase the risk of injury from thoracic disease and surgery compared to the right RLN.[15] Left RLN reinnervation is more demanding than on the right side because it requires greater energy expenditure for protein translation and transport from the cell body to the nerve growth cone.[16–19] Intra-axonal transport also implicates the reliance on rapid mitochondrial transport to areas along the axon to meet high energy demand.[20,21] This phenomenon would make the longer left RLN harder to regenerate after acute injury compared to the right.

Cases of idiopathic vocal fold paralysis and paralysis after short-term intubation for surgery also show similar discrepancies in incidence between sides.[22–25] These are states where the RLN extension and trajectory should not impact the rate of injury. This persistent sidedness

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suggests intrinsic neuromuscular differences between the left and right sides may also play a role in the frequency of paralysis.[25] In addition, previous findings indicate that intrinsic laryngeal muscles (ILM) exhibit different fiber-type compositions on the left versus the right side in equines.[10] Considering that targeted axon outgrowth to the ILMs requires a dialogue between the motoneurons and muscle fibers, an analysis of ILM gene expression may provide insights into the asymmetry of the larynx.[26] These changes in gene expression could also explain the sidedness of vocal fold paralysis. Moreover, organ asymmetry has already been identified in the diaphragm muscles and the phrenic nerves that innervate them.[27] If intrinsic neuromuscular differences in the larynx (aside from length) influence disparities in reinnervation between sides, these differences may be exploited as therapeutic targets.

Understanding that muscle type may vary from left to right, that neuromuscular factors aside from length may result in the increased incidence of left-sided vocal fold paralysis, and that signals from a muscle influence its innervation, the present work aims to investigate whether there are inherent asymmetric discrepancies in ILM basal gene expression related to neuromuscular innervation.[28–31] Such genes could contribute to the disparity in the incidence of left versus right vocal fold paralysis and would warrant further investigation relative to changes after nerve injury. This study evaluates distinct transcriptome sequences in the left versus right posterior cricoarytenoid (PCA), lateral (LTA), and medial thyroarytenoid (MTA) muscles of uninjured rats using recently developed techniques.[32]

In addition to the specific focus of this study, the differentially expressed genes identified in this research may also be useful as controls in future studies exploring laryngeal muscle function as it relates to other diseases associated with neurophysiological, myogenic, immune or hormonal control. This could include such disorders as Parkinson’s disease, dysphagia, vocal fold atrophy, presbylarynges, muscle tension dysphonia, and vocal fold paresis.[33–38]

MATERIALS AND METHODS

Rats and Housing

88 samples were obtained from 80 Sprague Dawley rats distributed in 12 groups according to age (adult and P15), sex (male and female), and type of muscle (PCA, LTA, and MTA). Tissue pooling is necessary to obtain sufficient total RNA. Young adult rats ($N = 24$) ~300 g required three pooled muscle bellies for sufficient RNA quantities; P15, post-neonatal rats 15 days after birth ($N = 56$) weighed ~30 g required five to seven pooled muscle bellies for consistent RNA recovery. For instance, three right PCA muscles were used for one right PCA sample. Muscles from nine adult females produced 54 muscles. These are pooled into three muscles per sample, resulting in 18 total samples. Muscles from 15 adult males produced 90 muscles. These were pooled into three muscles per sample, resulting in 30 samples. Unfortunately, three of these samples had an insufficient RIN and were removed for a total of 27 samples. Contrary to the adults, P15 animals required a minimum of five and optimally seven pooled muscles to yield enough RNA. Five muscles were used if technical difficulties prevented optimal dissection of the muscle. Muscles from 28 P15 females ultimately provided 20 samples. Similarly, muscles from 28 P15 males provided

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23 samples. A schematic representation of the experimental design is illustrated in Figure 1. This study was performed in compliance with the Public Health Service Policy on Humane Care and Use of Laboratory Animals, the National Institutes of Health Guide for the Care and Use of Laboratory Animals, the Animal Welfare Act (7 U.S.C. et seq.), and the ARRIVE guidelines. The Institutional Animal Care and Use Committee of Columbia University Medical Center approved the animal use protocol.

Surgical Procedure and Tissue Processing

All rats were euthanized via an overdose of ketamine and xylazine before the larynx was dissected. ILM isolation for RNA-Seq was performed as previously described.[33] Harvested larynxes were stabilized by briefly rinsing in 2 mL of RNA*later* (catalog #AM7021, Invitrogen, Waltham, MA, U.S.A.) prior to dissection of the muscle bellies using a Leica S8 APO compound microscope (Leica Microsystems, Buffalo Grove, IL). With micro-forceps and micro-scissors, the left and right PCA, LTA, and MTA were carefully excised and stored overnight at 4 C° in 1 mL RNA*later.* The next day, excess stabilizing reagent was removed, and the muscle bellies were placed into BioMasher III (Nippi, Inc., Japan) homogenization tubes according to age, sex, and muscle type (Figure 1). The pooled ILM muscle bellies were separated into different homogenization tubes were physically disrupted in liquid nitrogen before adding lysis buffer from the Aurum Total RNA Mini Kit (catalog #732-6820, Bio-Rad, Hercules, CA, U.S.A.). Further homogenization was conducted using a TissueRuptor II (catalog #9002755, Qiagen, MD). Tissue samples from distinct tubes underwent separate processing steps for RNA extraction. All proceeding steps were according to the commercial RNA isolation protocol. Total RNA was measured using a micro-volume spectrophotometer NanoDrop ONE (catalog #ND-ONE-W, Thermofisher, Wilmington, DE). Agilent RNA 6000 Pico Kit (Eukaryote Total RNA Pico, Agilent Technologies, Santa Clara, CA, U.S.A.) and Agilent 2100 bioanalyzer (Agilent Technologies, Santa Clara, CA, U.S.A.) were used to evaluate the RIN. The recovered RNA used in the subsequent sequencing steps reached a RIN value of 7.9 ± 0.4 ($N = 90$).

Library Preparation and RNA Sequencing

The RNA sequencing procedures were performed using a poly-A pull-down method to enrich mRNA from the total RNA samples. Library construction utilized TruSeq chemistry through the Illumina TruSeq Stranded mRNA kit and further sequencing was conducted using the Illumina NovaSeq 6000 (Illumina Inc., San Diego, CA, U.S.A.) desktop sequencer at the Columbia University Genome Center. Standard analytics included RTA (Illumina) for base-calling and bcl2fastq2 (version 2.19) for BCL to a fastq format conversion, along with adaptor trimming. Samples were multiplexed in each lane and pseudo-aligned to a kallisto (0.44.0) index. After quality control steps, the pseudo-aligned reads were mapped to the annotated *Rattus norvegicus* genome assembly Ensembl database.

Measured Expression Analysis

Differential gene expression analysis in the left versus the right respective ILM was performed through the EdgeR Bioconductor Package 3.32. Sample counts were normalized with qsmooth. Limma-zoom was used to evaluate the sample weights and downweigh the

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effect of sample outliers. A comprehensive list of differentially expressed genes (DEGs) was generated under three different conditions in the left versus the right comparison. 88 samples, not including the controls, were dichotomized into “left” vs. “right” groups according to the muscle, age, and gender of the rat. The total number of DEGs exhibited using False Discovery Rate (FDR)<0.05 and log fold change >|0.6| across all rat cohorts is displayed in Table 1. Raw data can be found in the Gene Expression Omnibus (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM6703793).

iPathwayGuide Analysis

Advaita Bioinformatics iPathwayGuide analysis identified significant genes based on a log fold change of expression with an absolute value of at least 0.6 and a 0.05 threshold for the p-value. Data were analyzed in the context of pathways obtained from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (Release 100.0+/11–12, Nov 21). Gene Ontologies were obtained from the Gene Ontology Consortium Database. Impact analyses for pathway analysis were evaluated by over-representation of differentially expressed genes in a given pathway and the perturbation of that pathway computed by propagating the measured expression changes across the pathway topology. Pathways from the P15 and adult datasets were compared to identified systematic commonalities.

RESULTS

Sequencing of the transcriptome across the ninety ILM samples resulted in a mean of 32.9 million mapped reads and a 79.1% mean mapping ratio, with no less than 70% sequence alignment in any sample. Principal component (PC) analysis by experimental condition (i.e., sex, age, ILM) indicated that PC1 and PC2 accounted for 98% variance. Samples were normalized, and then, using a mean-variance trend into precision weights, the effects of sample outliers were downweighed. Further quality control measures validated the differential control analysis (Figure 2A).

Overview of Differential Gene Analysis

Differential expression analysis confirmed asymmetric gene transcriptome profiles between the left-right ILMs according to age, sex, and muscle. Across the PCA and LTA experimental groups, DEGs showed only lower expression (downregulation) in the left compared to the right ILM (Supplemental Figure S1). In contrast, the MTA datasets detected DEGs with both lower expression (downregulation) and higher expression (upregulation) in the left versus right ILM (Supplemental Figure S1). At least 63% of DEGs in the P15 and 95% of DEGs in the adult MTA were higher (upregulated) on the left relative to the right side. Overall, adult males exhibited the most substantial number of DEGs in each muscle type, with comparatively few DEGs identified in adult females or at P15 in either sex (Figure 2B). Apart from the MTA, shared significant DEGs were not observed among the adult datasets (Figure 2C). Perturbation and overrepresentation analysis revealed enriched KEGG Pathways in the left versus right P15 male MTA, adult female PCA, and the adult male PCA, LTA, and MTA datasets (Tables 1, 2, 3, 4). While no shared DEGs were found between distinct ILMs across different ages and sexes, identified KEGG pathways were found to have similar molecular mechanisms. Gene Ontology (GO) and KEGG Pathway

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enrichment analysis of the significant left-right DEGs indicated that the implicated genes across experimental conditions mainly participate in neurophysiological and myogenic processes, inflammatory/immune response, and hormonal control (Figures 3 to 6). These findings varied according to age, sex, and distinctive ILM.

PCA—Across the global data, the PCA showed the fewest DEGs of any comparison (Figure 2B). GO Classifications and KEGG Pathways in the left-right adult male PCA indicated that DEGs were highly associated with inflammatory/immune response mechanisms, hormonal control, and neurophysiological processes (Figure 3, Table 1). Common KEGG pathways between the adult male PCA and MTA datasets showed considerable enrichment of DEGs involved in inflammatory/immune signaling (Table 1 and 3). The top pathways of the adult male included ‘antigen processing and presentation’ in addition to ‘Th1 and Th2 cell differentiation’, which are established inflammatory/immune response mechanisms (Table 3). Interestingly, differential expression analysis in the left-right adult male PCA did not implicate the involvement of myogenic processes in the PCA asymmetry. The adult female PCA only exhibited differential expression of a single mitochondrial iron ion transporter gene, Slc25a37, between sides and thus was not amenable to further GO or biological pathway analysis.

LTA—The adult male LTA dataset was the only condition to exhibit significant differential gene expression in the left compared to the right. GO and KEGG pathway enrichment analysis indicated that DEGs represented in the adult male LTA were highly associated with neurophysiological, myogenic, inflammatory, and hormonal regulation (Figure 4). Specifically, top GOs included the ‘semaphorin-plexin signaling pathway’, a hallmark of axon guidance, and the ‘postsynaptic neurotransmitter receptor cycle’ (Figure 4). Top myogenic factors and inflammatory response included ‘satellite cell activation involved in skeletal muscle regeneration’ and ‘negative regulation of necrotic cell death’, respectively (Figure 4). Top KEGG Pathways demonstrated highly associated DEGs involved in hormonal control, such as ‘growth hormone synthesis, secretion, and action’, and neurophysiological processes, such as the ‘calcium signaling pathway’ (Table 1).

MTA—Overall, the MTA showed the most DEGs of any condition (Figure 2B). DEGs in this muscle group were identified across sexes in both the adult and P15 datasets, although the high number of DEGs intersecting between the adult male and female MTA is most likely attributable to residual mucosa surrounding the muscle during dissection. GO Classification in the P15 and adult male and female MTA datasets demonstrated the enrichment of inflammatory, neurophysiological, myogenic, and hormonal processes (Figures 5 and 6). However, no common pathways that regulate hormonal control were found between the adult male and female datasets. Similar KEGG Pathways that direct hormonal control were identified between the adult and P15 male MTA (Tables 1 and 2). Top GO Classifications in the P15 male MTA included relevant myogenic and neurophysiological factors, such as ‘regulation of muscle adaptation’ and ‘inhibitory chemical synaptic transmission in response to denervation’ (Figure 6A). Relevant neurological pathways across the adult male, adult female, and P15 male MTA included ‘neuroactive-ligand receptor interaction’ and ‘adrenergic signaling in cardiomyocytes’, which may be similar to pathways in striated

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muscles that are less characterized (Tables 1 and 2). Considering the difficulty of cleanly dissecting the MTA from the mucosa, some of the findings in the MTA may be contaminated by the mucosa. Additional experiments are warranted to confirm whether the observed inflammatory processes and hormonal control are specific to the MTA.

Age-related differences—The P15 datasets showed less pronounced left-right differences in gene expression. This phenomenon may be attributed to the delay of sexual maturation, which occurs approximately six weeks after rat birth. P15 animals are considered sexually immature pups as a result.[39] Unlike the adults, the P15 PCA and LTA datasets showed no lateral differences (Figure 2B). GO and KEGG Pathway enrichment analysis revealed that only a fraction of genes in the male and female P15 MTA datasets participate in inflammatory mechanisms (Figure 6). In contrast, immune response regulators predominated in the adult experimental conditions (Figures 3 to 5).

Sex-specific changes—Sex-specific differences were apparent through the absence of shared DEGs, GOs, and KEGG Pathways between ILMs of the same age and muscle type. Given the lack of significant DEGs across the female PCA and LTA datasets, the adult and P15 MTA datasets allowed for sex-related assessments (Figures 2B and C). Further analysis revealed fewer GO and KEGG Pathways related to hormonal control in the adult and P15 female MTA datasets than their male counterparts (Figures 5 and 6).

DISCUSSION

The left-right ILM transcriptome profile indicates that laterality is influenced by age, sex, and tissue-specific factors. Reviewing the KEGG Pathways and GO Classifications of DEGs across all experimental conditions allows for identifying genes that may contribute to laryngeal innervation and reinnervation. The results of this study indicated differences in transcriptional regulation between the left-right abductor (PCA) and adductor (LTA and MTA) muscles. Of the 774 total significant DEGs across all experimental conditions, 97.2% were identified within the adductor muscle group. While no shared DEGs were found between distinct ILMs across different ages and sexes, the enriched KEGG pathways indicated that the implicated genes possess similar molecular mechanisms. This suggests diverse genetic differences influence similar physiological changes that may impact the laterality of innervation in the healthy state.

The differential expression analysis revealed four molecular mechanisms that may influence genetic asymmetry within the uninjured ILMs: myogenic and neurophysiological factors, hormonal control, and inflammatory response (Figures 3 to 6, Tables 1 to 2). The expression of genes involved in these processes varied per experimental group. While genes involved in hormonal control, neurophysiology, and myogenesis showed mostly lower expression in the left compared to right muscles, most genes implicated in inflammatory or immune response were found to have higher expression on the left side. Genes linked to the inflammatory response exhibited the most significant overall expression levels in the global data. This finding corresponds with the reliance on immune response and hormone signals to promote peripheral nerve regeneration after injury. In the former case, functional recovery of the peripheral nerve tissue is facilitated by inflammatory-related genes to

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the innate immune response.[40] Neurotrophic and guidance factors are upregulated during nerve regeneration, while myelin-associated inhibitors are cleared to promote distal neurite outgrowth to the original targets.[41] In the latter event, recent studies have evidenced hormonal control as an essential component of muscle reinnervation after nerve injury. More specifically, growth hormone treatment in the rat forelimb was associated with significantly attenuated muscle atrophy and enhanced nerve regeneration, muscle reinnervation, and motor function.[42] Thyroid hormone and its receptors, localized in Schwann cells, are also critical for promoting nerve regeneration.[43] The absence of DEGs related to inflammation in P15 muscles may reflect the development of the immune system between P15 and adulthood.[44] Indeed, this phenomenon would agree with age-dependent and sex-related studies concerning musculature hormone control, though further analysis is needed to justify this association.[11,13,45–48]

While the increased risk of vocal fold paralysis due to aging is well-supported, clinical evidence has not conclusively established sex as a risk factor.[25,49–51] Still, the results of this study and emerging evidence linking sex hormones to peripheral nerve function suggest a potential influence of sex in RLN regeneration/survival and subsequent paralysis[47–54].[52–59] Earlier studies noting sex hormone receptors at the vocal fold level and their influence on musculature and neurotrophic factors support this connection.[54–56,59–62] Moreover, findings showcasing estrogen and progesterone's modulation of nerve growth factors in peripheral organs may explain the left-right ILM gene discrepancies between male and female adults—although this should be assessed in future studies for validation.[56]

This preliminary study aimed to investigate genetic asymmetries related to left-right ILM innervation in an uninjured model. While musculoskeletal genes show more similarities than differences between the left and right ILMs, the discovery of asymmetric expression of significant DEGs with neuromuscular functions indicates these differences may have meaningful physiological implications. It also implicates genetic protein expression in the musculoskeletal system as one of many determinants of laterality outcomes. Of particular interest are genes such as Sema3d and Plxnb1, which are known to play a significant role in axonal growth cone guidance and target innervation.[63–65] Further investigation is needed to confirm if the genetic variations identified in the study between the left and right ILMs contribute to the more frequent occurrence of left-sided paralysis. This finding would challenge the conventional belief that asymmetry in paralysis rates is solely due to differences in nerve length between the left and right RLN. Genetic insights from this and future studies will aid in assessing gene signatures, guiding therapeutic development to improve ILM reinnervation and restore normal vocal fold function after RLN injury.

Potential limitations of this study include (1) examining the ILMs through bulk rather than single-cell RNA-seq, (2) the risk of including mucosa or fibers from adjacent muscles during laryngeal muscle microdissection, and (3) the tissue pooling size. Differences in DEGs related to laterality were more noticeable among gene sets than at the individual gene level. Identifying subtle differences in gene expression will require validation through functional assays like RT-PCR. Inaccurate muscle dissection could confound the sequencing results with that of adjacent muscle or mucosa contamination. While confident in our microsurgical techniques, this study acknowledges the inherent difficulty in isolating the left and right

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PCA at the midline raphe and removing all remnants of mucosa from the left and right MTA. Normalization and downweighing of sample outliers were performed during the quality control step to mitigate the impact of this issue. The pooling strategy was employed to systematically acquire sufficient high-quality RNA by aggregating muscle samples from three distinct rats. However, the influence of multiple transcripts per sample may perturb sequencing results if there is transcriptome overrepresentation in a given sample. Increasing the number of biological replicates or the pooled sample size to at least eight tissues to minimize the false discovery rate effectively reduces pooling bias.[32,66,67] Further details regarding the experimental design technical limitations are outlined in Kemfack et al. (2022).[64]

CONCLUSION

Differential expression analysis of the left-right ILM transcriptome identified age, sex, and muscle-dependent differences in the healthy state. Several common processes implicated in the laterality of ILM gene expression have been shown to influence innervation, These factors include neurophysiological and myogenic processes, inflammatory response, and hormonal control. The genes of interest identified in this study will be evaluated after RLN injury and in models of reinnervation to study their impact on RLN reinnervation. The findings from these investigations may be also critical to developing enhanced treatments to address several neuromuscular degenerative diseases and other muscle-related pathologies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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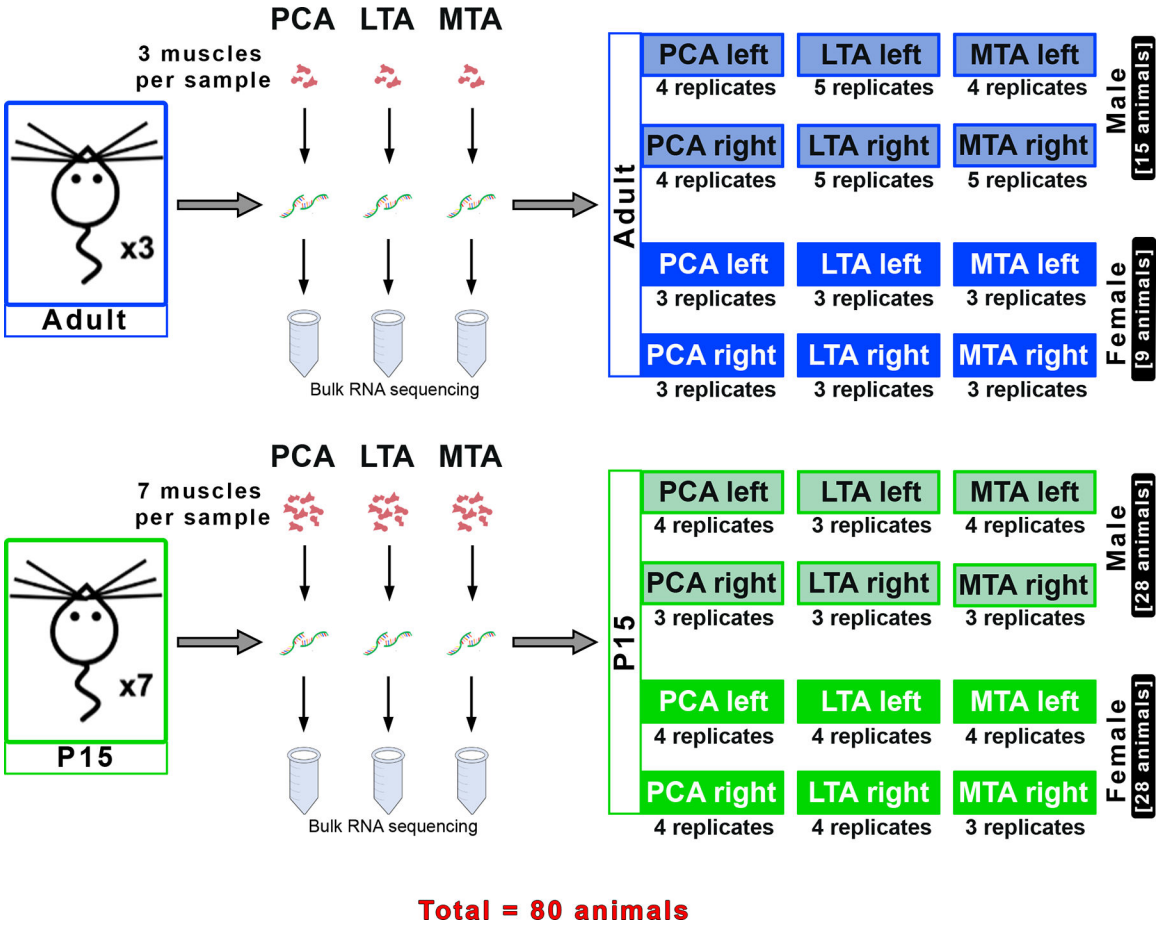


Figure 1.

A schematic diagram illustrating the experimental design for high-throughput RNA-Sequencing (RNA-Seq) of the distinctive intrinsic laryngeal muscles.

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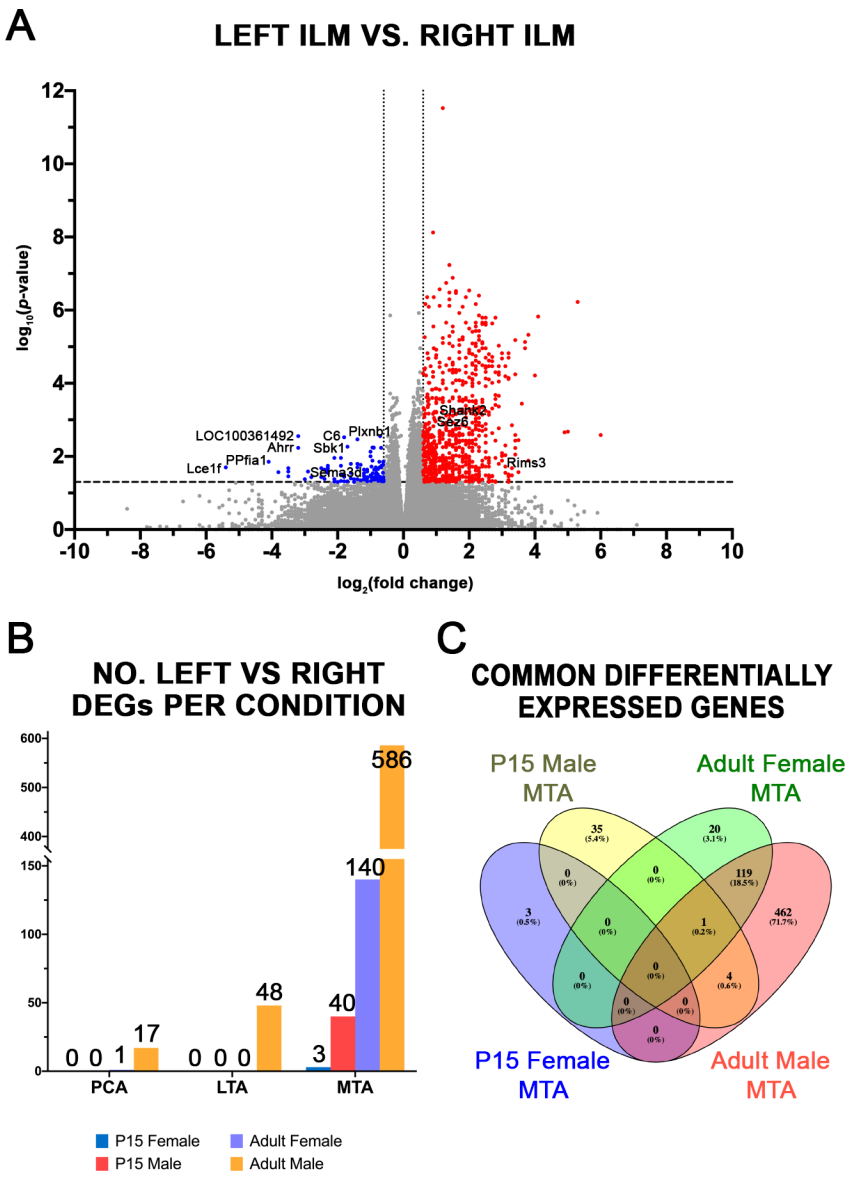


Figure 2.

Global differential expression analysis summary comparing gene regulation in the left versus right ILMs. (A) Volcano Plot showing statistically significant genes and their fold change in gene expression across the P15 and adult PCA, LTA, and MTA muscles in both male and female rats. Downregulated differentially expressed genes (DEGs) on the left compared to the right ILMs are represented in blue, while upregulated DEGs are depicted in red. DEGs that were not significant (p-value > 0.05; log fold change > |0.6|) are shown in gray. (B) Number of DEGs in the left compared to the right across each experimental condition. (C) Venn diagram indicating the number of genes that overlapped across all MTA datasets. Common DEGs across the global data intersected three datasets in the MTA muscle.

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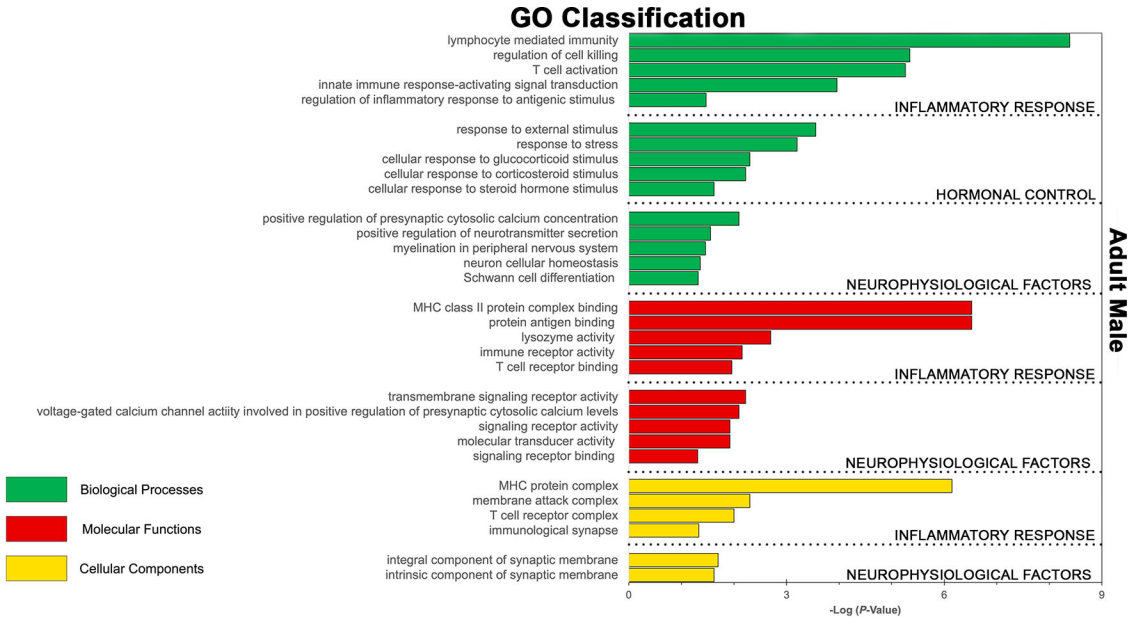


Figure 3.

Key Gene Ontology pathways enriched in the PCA muscle. Top Gene Ontologies (GOs) enriched in the adult male left vs. right PCA muscle are classified by inflammatory, hormonal, and neurophysiological sub-types in order of statistical significance. Biological processes are shown in green bars; molecular functions are represented in red bars, and cellular components are depicted in yellow bars.

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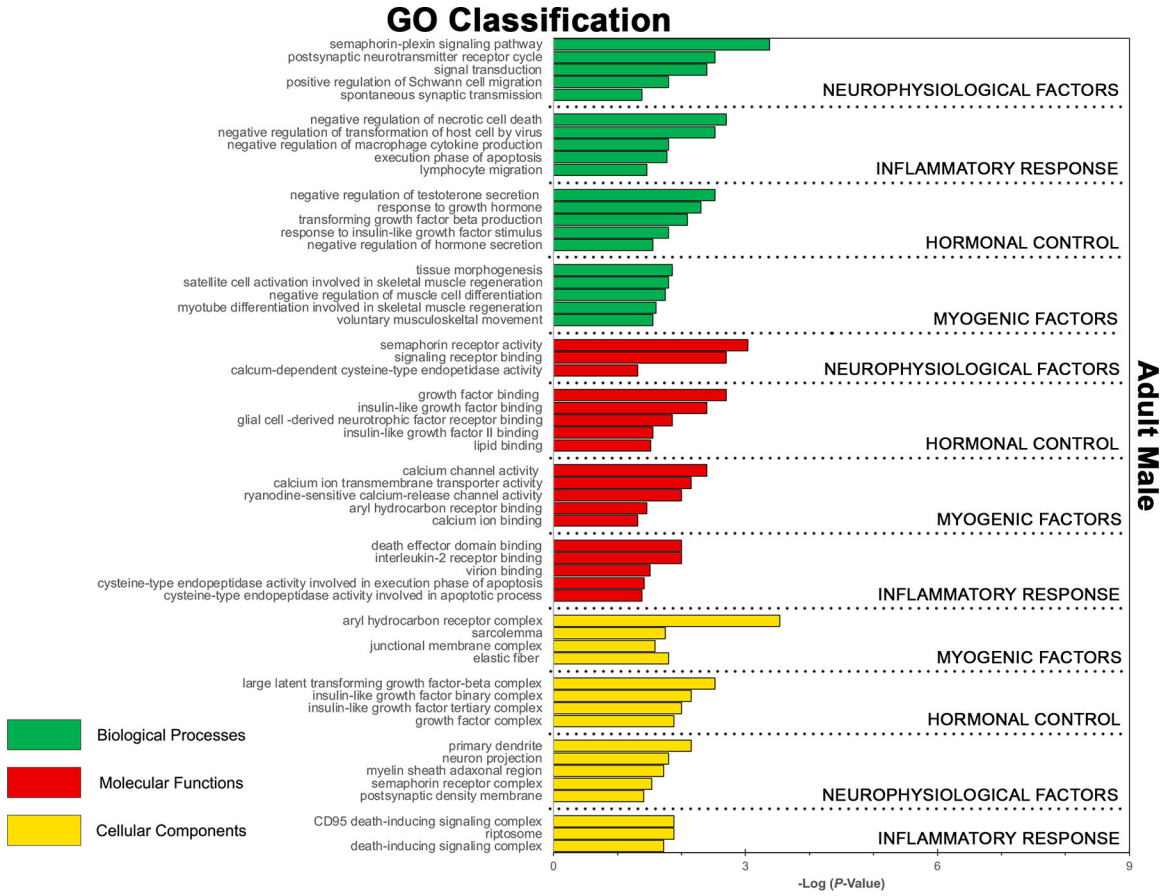


Figure 4.

Key Gene Ontology pathways enriched in the LTA muscle. Top Gene Ontologies (GOs) enriched in the adult male left vs. right LTA muscle are classified by neurophysiological, inflammatory, hormonal, and myogenic sub-types in order of statistical significance. Biological processes are shown in green bars; molecular functions are represented in red bars, and cellular components are depicted in yellow bars.

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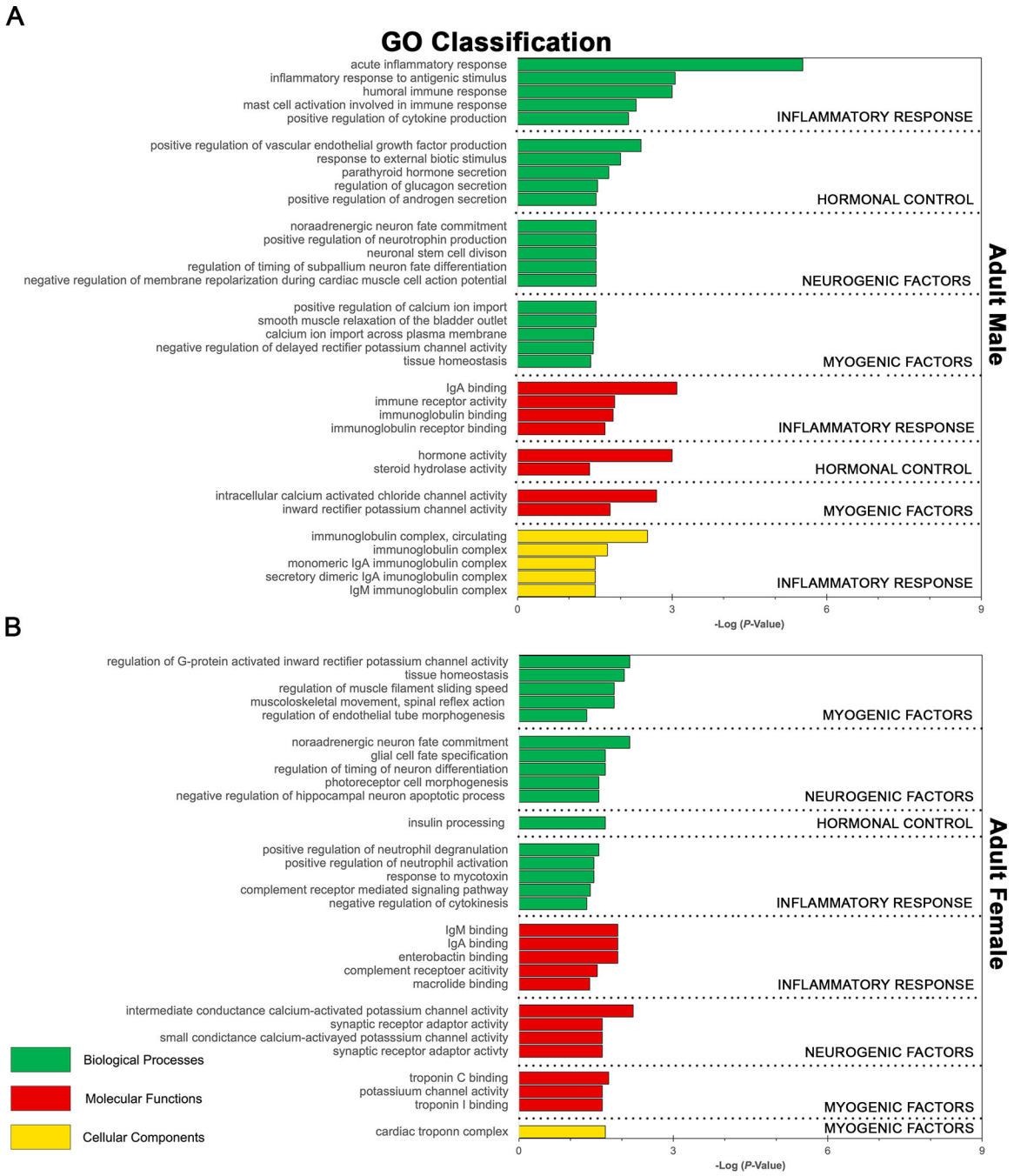


Figure 5.

Key Gene Ontology pathways enriched in the MTA muscle. Top Gene Ontologies (GOs) enriched in the adult male left vs. right MTA are classified by inflammatory, hormonal, neurophysiological, and myogenic factors in order of statistical significance. (B) Top Gene Ontologies (GOs) enriched in the adult female left vs. right MTA are classified by myogenic, neurophysiological, hormonal, and inflammatory factors in order of statistical significance. Biological processes are shown in green bars; molecular functions are represented in red bars, and cellular components are depicted in yellow bars.

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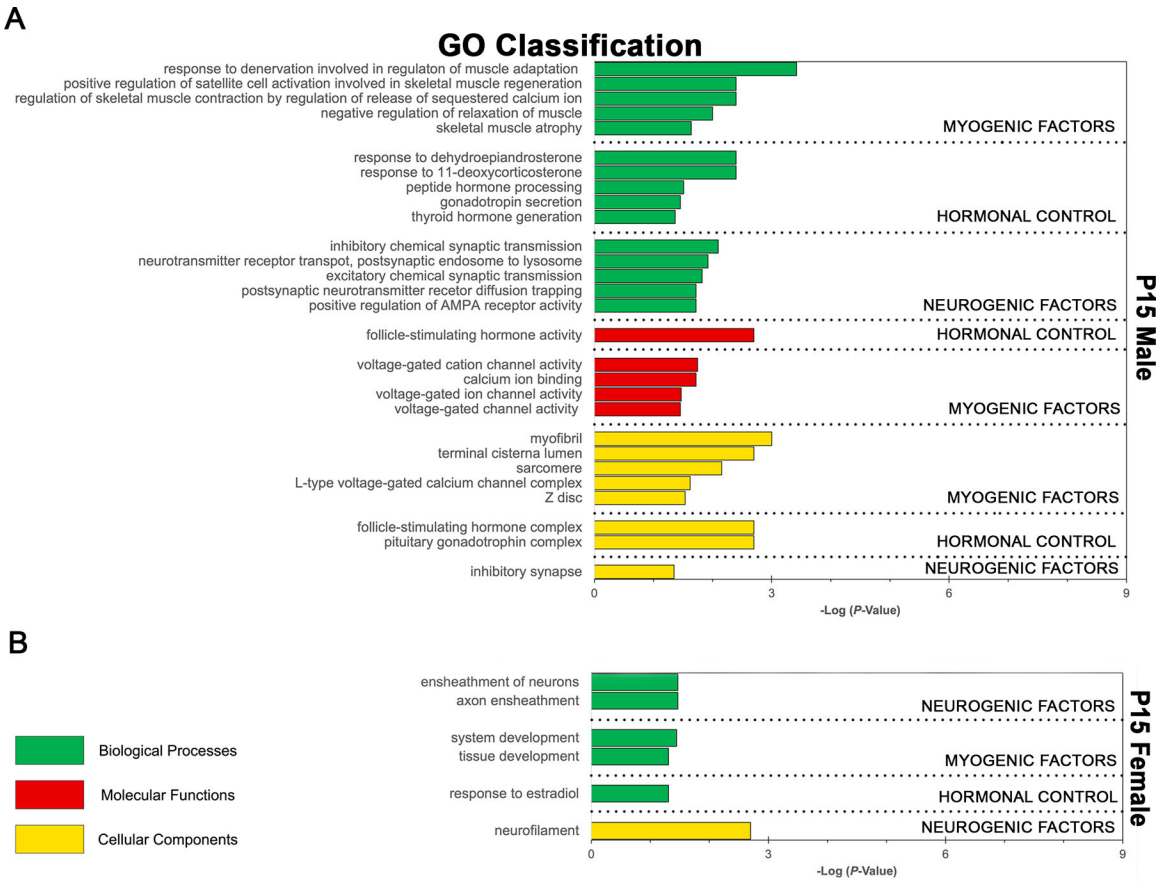


Figure 6.

(A) Top Gene Ontologies (GOs) enriched in the P15 male left vs. right MTA muscle are classified by myogenic, hormonal, and neurophysiological sub-types in order of statistical significance. (B) Top Gene Ontologies (GOs) enriched in the P15 female left vs. right MTA muscle are classified by myogenic, hormonal, and neurophysiological sub-types in order of statistical significance. Biological processes are shown in green bars; molecular functions are represented in red bars, and cellular components are depicted in yellow bars.

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Table 1.

Kyoto Encyclopedia of Genes and Genomes pathway enrichment analysis of Differentially Expressed Genes associated with innervation mechanisms in the Adult Male PCA, LTA, and MTA datasets.

Enriched KEGG Pathways	p-value	genes	List of genes
PCA Adult Male			
Neuroactive ligand-receptor interaction	0.161	**1** / 210	Cnr1
LTA Adult Male			
Calcium signaling pathway	0.058	**3** / 201	Mast1r, Orai2, Ryr3
Growth hormone synthesis, secretion and action	0.195	**1** / 104	Igfbp3
Oxytocin signaling pathway	0.246	**1** / 137	Ryr3
Axon guidance	0.309	**2** / 171	Sema3d, Plxnb1
Neurotrophin signaling pathway	0.515	**1** / 113	Irak3
Serotonergic synapse	0.563	**1** / 91	Ptgs1
Pathways of neurodegeneration – multiple diseases	0.709	**2** / 416	Casp8, Ryr3
MTA Adult Male			
Neuroactive ligand-receptor interaction	0.015	**11** / 210	Kng1, Fpr-rs3, C3, Nts, Iapp, Nmur1, Adora2b, Lhb, Adm2, Calcb, Ptgdrl
Pathways of neurodegeneration - multiple diseases	0.137	**12** / 416	Dnah1, Dnah6, Dnali1, Dnah2, Dnah12, Hap1, Dnai2, Dnah9, Dnai1, Dnah5, Dnah7, Il1b
cGMP-PKG signaling pathway	0.183	**2** / 148	Kng1, Pik3cg
Renin secretion	0.211	**3** / 63	Clca4, Clca4l, Clca1
VEGF signaling pathway	0.225	**2** / 56	Pla2g4c, Rac2
Glutamatergic synapse	0.344	**3** / 101	Clca4, Clca4l, Clca1
Serotonergic synapse	0.383	**2** / 91	Pla2g4c, Cyp2d3
Taste transduction	0.414	**1** / 47	Trpm5
Long-term depression	0.418	**1** / 57	Pla2g4c
Morphine addiction	0.453	**1** / 76	Grk3
GABAergic synapse	0.459	**2** / 76	Slc38a5, Hap1
Retrograde endocannabinoid signaling	0.502	**1** / 123	Faah
Alzheimer disease	0.505	**1** / 334	Il1b
Vascular smooth muscle contraction	0.687	**5** / 118	Adora2b, Pls2g4c, Calcb, Ramp3, Adm2
Synaptic vesicle cycle	0.729	**2** / 91	Atp6v1b1, Atp6v1c2
Olfactory transduction	0.894	**1** / 39	Grk3

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Enriched KEGG Pathways	p-value	genes	List of genes
PCA Adult Male			
Calcium signaling pathway	0.949	**3** / 201	Adora2b, Gna14, Vegfd
Neurotrophin signaling pathway	0.953	**1** / 113	Tp73
Cholinergic synapse	0.982	**1** / 103	Pik3cg
Axon guidance	0.986	**1** / 171	Rac2
Adrenergic signaling in cardiomyocytes	0.989	**1** / 139	Pik3cg

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Table 2.

Kyoto Encyclopedia of Genes and Genomes pathway enrichment analysis of Differentially Expressed Genes associated with innervation mechanisms in the Adult Female MTA and P15 Male MTA datasets.

Enriched KEGG Pathways	p-value	genes	List of genes
MTA Adult Female			
Huntington disease	0.010	**7** / 264	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Dnali1
Amyotrophic lateral sclerosis	0.022	**7** / 314	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Dnali1
Pathways of neurodegeneration - multiple diseases	0.033	**8** / 416	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Wnt5b, Dnali1
Neuroactive ligand-receptor interaction	0.207	**3** / 210	Ptgdrl, Fpr-rs3, Rxfp2
Alcoholic liver disease	0.340	**1** / 113	Scd3
AMPK signaling pathway	0.372	**1** / 114	Scd3
Relaxin signaling pathway	0.401	**1** / 115	Rxfp2
Glutamatergic synapse	0.421	**1** / 101	Homer2
Adrenergic signaling in cardiomyocytes	0.425	**1** / 139	Tnnt2
mTOR signaling pathway	0.492	**2** / 145	Atp6v1c2, Wnt5b
FoxO signaling pathway	0.778	**1** / 115	Homer2
Axon guidance	0.887	**1** / 171	Wnt5b
Wnt signaling pathway	0.893	**1** / 149	Wnt5b
Alzheimer disease	0.990	**1** / 334	Wnt5b
MTA P15 Male			
Thyroid hormone synthesis	0.074	**1** / 67	Cga
Adrenergic signaling in cardiomyocytes	0.142	**2** / 139	Cacng8, Ppp1r1a
Neuroactive ligand-receptor interaction	0.210	**1** / 210	Cga
Calcium signaling pathway	0.261	**1** / 201	Casq1
Parathyroid hormone synthesis, secretion and action	0.404	**1** / 92	Runx2
Wnt signaling pathway	0.628	**1** / 149	Notum

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Table 3.

Top enriched KEGG Pathways of differentially expressed genes (DEGs) in the Adult PCA, LTA, and MTA datasets.

Enriched KEEG Pathways	p-value	genes	List of genes
PCA Adult Male			
Asthma	2.93E-06	**3** / 16	RT1-Bb, RT1-Ba, RT1-Da
Systemic lupus erythematosus	3.19E-06	**4** / 59	C6, RT1-Bb, RT1-Ba, RT1-Da
Graft-versus-host disease	3.61E-06	**4** / 35	RT1-Bb, Klrd1, RT1-Ba, RT1-Da
Antigen processing and presentation	1.46E-05	**4** / 62	RT1-Bb, Klrd1, RT1-Ba, RT1-Da
Intestinal immune network for IgA production	1.79E-05	**3** / 33	RT1-Bb, RT1-Ba, RT1-Da
Inflammatory bowel disease	2.02E-05	**3** / 46	RT1-Bb, RT1-Ba, RT1-Da
Allograft rejection	2.31E-05	**3** / 35	RT1-Bb, RT1-Ba, RT1-Da
Type I diabetes mellitus	2.35E-05	**3** / 42	RT1-Bb, RT1-Ba, RT1-Da
Autoimmune thyroid disease	2.75E-05	**3** / 39	RT1-Bb, RT1-Ba, RT1-Da
Leishmaniasis	5.77E-05	**3** / 62	RT1-Bb, RT1-Ba, RT1-Da
Staphylococcus aureus infection	6.98E-05	**3** / 57	RT1-Bb, RT1-Ba, RT1-Da
Viral myocarditis	1.00E-04	**3** / 65	RT1-Bb, RT1-Ba, RT1-Da
Th17 cell differentiation	1.15E-04	**3** / 93	RT1-Bb, RT1-Ba, RT1-Da
Rheumatoid arthritis	1.63E-04	**3** / 76	RT1-Bb, RT1-Ba, RT1-Da
Th1 and Th2 cell differentiation	1.75E-04	**3** / 78	RT1-Bb, RT1-Ba, RT1-Da
Toxoplasmosis	2.75E-04	**3** / 100	RT1-Bb, RT1-Ba, RT1-Da
Tuberculosis	5.48E-04	**3** / 147	RT1-Bb, RT1-Ba, RT1-Da
Influenza A	6.82E-04	**3** / 133	RT1-Bb, RT1-Ba, RT1-Da
Epstein-Barr virus infection	0.001	**3** / 189	RT1-Bb, RT1-Ba, RT1-Da
Human T-cell leukemia virus 1 infection	0.002	**3** / 212	RT1-Bb, RT1-Ba, RT1-Da
Retrograde endocannabinoid signaling	0.004	**1** / 123	Cnr1
Herpes simplex virus 1 infection	0.005	**3** / 282	RT1-Bb, RT1-Ba, RT1-Da
Coronavirus disease - COVID-19	0.006	**1** / 208	C6
LTA Adult Male			
Viral myocarditis	0.024	**1** / 65	Casp8
Transcriptional misregulation in cancer	0.027	**1** / 153	Igfbp3
MTA Adult Male			

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Enriched KEEG Pathways	p-value	genes	List of genes
PCA Adult Male			
Neuroactive ligand-receptor interaction	0.015	**11** / 210	Kng1, Fpr-rs3, C3, Nts, Iapp, Nmur1, Adora2b, Lhb, Adm2, Calcb, Ptgdrl
IL-17 signaling pathway	0.020	**7** / 79	Il25, Ccl20, Lcn2, Il17rb, Mapk15, Cxcl1, Il1b
Rheumatoid arthritis	0.024	**7** / 76	Atp6v1c2, Tnfrsf11a, Ccl20, Atp6v1b1, Cxcl1, Il1b, RT!-DOb
Maturity onset diabetes of the young	0.025	**3** / 13	Hnf1b, Iapp, Mafa
Staphylococcus aureus infection	0.035	**6** / 57	Cfi, C3, Krt18, Fpr-rs3, Krt31, RT1-DOb
MTA Adult Female			
Huntington disease	0.010	**7** / 264	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Dnali1
Amyotrophic lateral sclerosis	0.022	**7** / 314	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Dnali1
Pathways of neurodegeneration - multiple diseases	0.033	**8** / 416	Dnai2, Dnah9, Dnah12, Dnah1, Dnah6, Dnai1, Wnt5b, Dnali1

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Table 4.

Top enriched Kyoto Encyclopedia of Genes and Genomes and Differentially Expressed Genes in the P15 Male MTA datasets.

Enriched KEGG Pathways	p-value	genes	List of genes
MTA P15 Male			
Arrhythmogenic right ventricular cardiomyopathy	0.007	**2** / 70	Cacng8, Actn3
Ovarian steroidogenesis	0.029	**1** / 43	Cga
Transcriptional misregulation in cancer	0.030	**1** / 153	Runx2
Autoimmune thyroid disease	0.042	**1** / 39	Cga